

DRP F26 - Coxeter Group

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Contents

1 Introduction 2

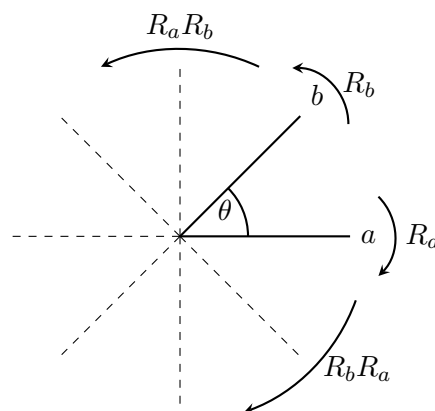
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1 Introduction

Def: Group

A **group** $(G, \bullet)$ where G is a set and $\bullet : G \times G \rightarrow G$ is a binary operator on the set, and it follows that:

- i) (Associativity): $\forall a, b, c \in G, (a \bullet b) \bullet c = a \bullet (b \bullet c)$
- ii) (Identity element): $\exists 1 \in G : \forall a \in G a \bullet 1 = 1 \bullet a = a$
- iii) (Inverse element): $\forall a \in G \exists a^{-1} \in G : a \bullet a^{-1} = a^{-1} \bullet a = 1$



Conjecture:

If mirrors a and b are at a $\theta = \frac{2\pi}{n}$ angle, the group generated by R_a, R_b is finite.

Idea:

- If $\theta \neq \frac{2\pi}{n}$ for any $n \in \mathbb{N}$, we don't get a finite partition of the planes.
- There is some relation between transformations generated by reflections and chambers.

Conjecture:

$$|G| = \text{number of chambers.}$$

Note

Root vectors...

$$R_a = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{aligned} R_{\vec{v}} &= \vec{v} - r(\vec{v} \cdot \vec{w}) \vec{w} \\ &= \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} - 2(-v_1 \sin \theta + v_2 \cos \theta) \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \end{aligned}$$

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